

PHAM THANH LAM

AI Engineer | Backend Developer (LLM Applications, Agentic Systems)
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PROFESSIONAL SUMMARY

Backend developer with 2+ years of experience building production REST APIs (.NET 8, Spring Boot, Go), now focused on LLM-powered applications. Built a Vietnamese personal-finance AI assistant (LangGraph agent, function calling, guardrails, human-in-the-loop confirmation, long-term memory, automated evaluation) and a RAG assistant for Discord communities on pgvector. Strong foundation in Clean Architecture, PostgreSQL/SQL Server, Redis, authentication and observability, applied to shipping reliable, safe AI features.

EDUCATION

Thuy Loi University 2021 – 2025
Bachelor of Information Technology Ha Noi

WORK EXPERIENCE

VSF 05/2026 – Present
AI / Backend Engineer (Probation) Ha Noi

- Built **Spectra**, a Vietnamese personal-finance AI assistant: a LangGraph agent with function calling, guardrails, long-term memory and an automated evaluation pipeline.
- Developing **PaySplit**, an AI OCR bill-splitting and settlement fintech app: Redis-backed session auth, SePay bank-transfer settlement and realtime group sync (SSE).
- Worked with spec-driven development, pull-request code reviews and AI-assisted engineering workflows (Claude Code).

Zamiga 05/2025 – 03/2026
Back-End Developer Ha Noi

- Designed and implemented scalable RESTful APIs using .NET 8 following Clean Architecture principles.
- Developed dynamic reporting and dashboard modules with Excel import/export integration.
- Optimized complex SQL queries to improve response time and data processing performance.
- Implemented JWT-based authentication and role-based authorization for secure API access.
- Collaborated in Agile teams (sprint planning, code reviews, daily stand-ups).

SFT Global 05/2024 – 04/2025
Backend-Focused Fullstack Developer (Intern → Developer) Ha Noi

- Built and maintained RESTful APIs using .NET and Spring Boot.
- Integrated Elasticsearch for real-time document search and indexing.
- Developed document processing features including watermark integration for content protection.
- Designed and optimized SQL queries for high-performance data retrieval; built Vue.js modules on top of these APIs.

AI PROJECT EXPERIENCE

Spectra – Personal Finance AI Assistant 05/2026 – 06/2026
AI Engineer / Main Developer (team of 2, VSF) github.com/lamphamaudio/personal-finance-ai-assistant

Technologies: Python, FastAPI, LangGraph, OpenAI / Gemini, PostgreSQL (Supabase), React + Vite, LangSmith, OpenTelemetry, pytest, Docker

- Agent orchestration:** Built a Vietnamese financial chatbot as a LangGraph state machine — guard, planner, tool executor, synthesis and output check — with conditional routing, multi-intent handling and clarification flows.
- Tool calling:** Designed the function-calling tool layer (spending analysis, budgets, savings goals, anomaly explanation, balance forecast, financial health score) so every answer is grounded in real transaction data.
- Safety:** Layered guardrails for prompt injection, out-of-scope and investment-advice requests, tool-argument validation and PII redaction on output; write actions require user confirmation and are audited.
- Memory:** Long-term user memory (preferences, category rules learned from corrections) with sensitive-data filtering, plus persistent chat sessions.
- Evaluation:** Built an automated evaluation pipeline covering intent classification, tool selection, confirmation flow and answer quality via LLM-as-a-Judge, run as a regression suite alongside unit tests.
- Observability:** Hierarchical tracing and structured logging for LLM and tool calls (LangSmith, OpenTelemetry) with payload redaction; migrated storage to Supabase Postgres and built a Bank Simulator service with SSO login.

KingModerator (A20) – RAG Assistant for Discord Communities 04/2026 – 05/2026
Fullstack AI Developer (team of 3) github.com/lamphamaudio/A20-App-131

Technologies: Python, FastAPI, PostgreSQL + pgvector, OpenAI, RAG, discord.py, React + TypeScript + Vite, Alembic, Braintrust, Docker

- Knowledge-first moderation platform: documents are uploaded, reviewed and published as knowledge-base versions, and the Discord bot may only answer from the active version — keeping replies grounded and rollback-able.

- Built the RAG retrieval path: document ingestion and semantic chunking with overlap, embeddings in pgvector, and grounded answer generation with source tracing per chunk.
- Implemented a hybrid content guard (regex first, LLM classifier second, behind a feature flag for cost control) detecting toxic, spam, off-topic and prompt-injection messages, with a moderation queue and remote message deletion on Discord.
- Built the help-center assistant end to end (article models, backend agent, admin UI, chat drawer) and covered its guardrails with tests.
- Developed the moderator dashboard in React/TypeScript: moderation queue, retrieval traces, AI suggestions, reply templates and markdown reply previews; fixed CSRF handling and session caching in authentication.

PaySplit – AI Bill Splitting & Settlement App

08/2026 – 09/2026

Backend & Mobile Developer (team of 3, VSF)

github.com/FintechVSF-Tranning

Technologies: Go, chi, PostgreSQL, sqlc, Redis, River queue, SSE, LlamaExtract OCR, VietQR, SePay, Firebase FCM, Flutter (Riverpod, Dio), Docker, Caddy

- Mobile fintech app that scans receipts with AI OCR, splits items (VAT, fees, discounts) among group members and settles debts through dynamic VietQR payments.
- Contributed to the bill module around the asynchronous OCR worker (receipt → LLM extraction → normalized items) and fixed item-level discount allocation and rounding edge cases in the split engine.
- Replaced JWT + Postgres sessions with opaque credentials backed by Redis (instant revocation, role cache, eviction-policy handling) and wrote the design rationale document.
- Built automatic bank-transfer settlement via SePay webhooks that matches incoming transactions to debts, and a realtime group sync over Server-Sent Events with delta catch-up.
- Implemented group management, pagination and debt-reminder APIs; built Flutter group hub, settlement and authentication screens on a feature-first Clean Architecture.

OTHER PROJECT EXPERIENCE

CQS

2025 – 2026

Back-End Developer

Technologies: .NET 8, .NET Framework 4.7, SQL Server, OpenID Connect (OIDC)

- Designed centralized authentication with OpenID Connect for Single Sign-On and secure role-based authorization.
- Integrated legacy .NET Framework 4.7 modules into the .NET 8 ecosystem; ensured transactional consistency in SQL Server.

Legacy System Migration – .NET 4.7.2 to .NET 8

2025 – 2026

Backend Developer

Technologies: .NET 8, .NET Framework 4.7.2, SQL Server

- Migrated a legacy application to .NET 8 with Clean Architecture; updated DI, middleware and authentication while keeping backward compatibility during rollout.

Document Management System

06/2024 – 10/2024

Backend Developer

Technologies: .NET 8, SQL Server

- Designed document lifecycle APIs (view, download, print, watermark).

SMS Service System

11/2024 – 02/2025

Backend Developer

Technologies: Spring Boot, SQL Server, Angular

- Designed message storage by customer type; built bulk Excel upload and secured image upload/download APIs.

TECHNICAL SKILLS

- **AI / LLM:** LangGraph, RAG (chunking, embeddings, pgvector), OpenAI & Gemini APIs, function/tool calling, prompt engineering, guardrails & PII redaction, agent memory, LLM-as-a-Judge evaluation, LangSmith, OpenTelemetry, OCR (LlamaExtract), scikit-learn
- **Languages:** Python, C#, Go, Java, JavaScript, Dart
- **Backend:** FastAPI, .NET 8 / ASP.NET Core, Spring Boot, Go (chi), RESTful API, SSE, Clean Architecture, OIDC / JWT
- **Data & Infra:** PostgreSQL (Supabase), SQL Server, Redis, MySQL, Firebase, Docker, background job queues
- **Mobile:** Flutter
- **Tools:** Git, GitHub, GitLab, SVN, Postman, Visual Studio / VS Code, Claude Code, Figma, Draw.io
- **Languages spoken:** Vietnamese (native), English (reading and writing technical documents)

HONORS & AWARDS

Third Prize, Provincial Excellent Student Contest in Mathematics

2019